5.Write a Python function to analyze the histogram of the given input image based on color levels using Open CV.

CODE:

```
import cv2

import matplotlib.pyplot as plt

img = cv2.imread(r"C:\Users\SUTHISH\Downloads\images.jpg")

colors = ('b', 'g', 'r')

for i, color in enumerate(colors):

    hist = cv2.calcHist([img], [i], None, [256], [0, 256])

    plt.plot(hist, color=color)

plt.title('Color Histogram')

plt.xlabel('Pixel Intensity')

plt.ylabel('Frequency')

plt.show()
```

OUTPUT:

